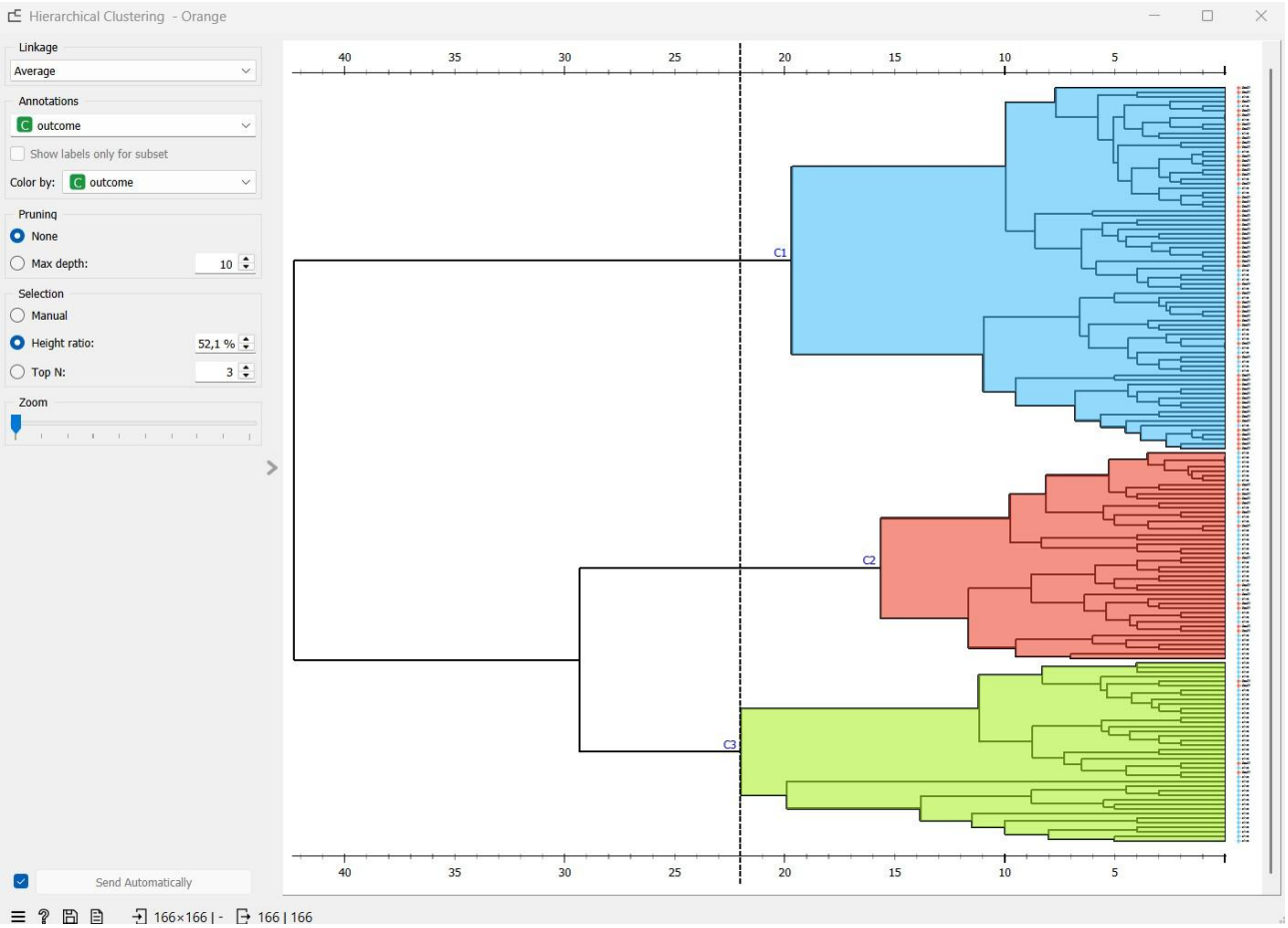
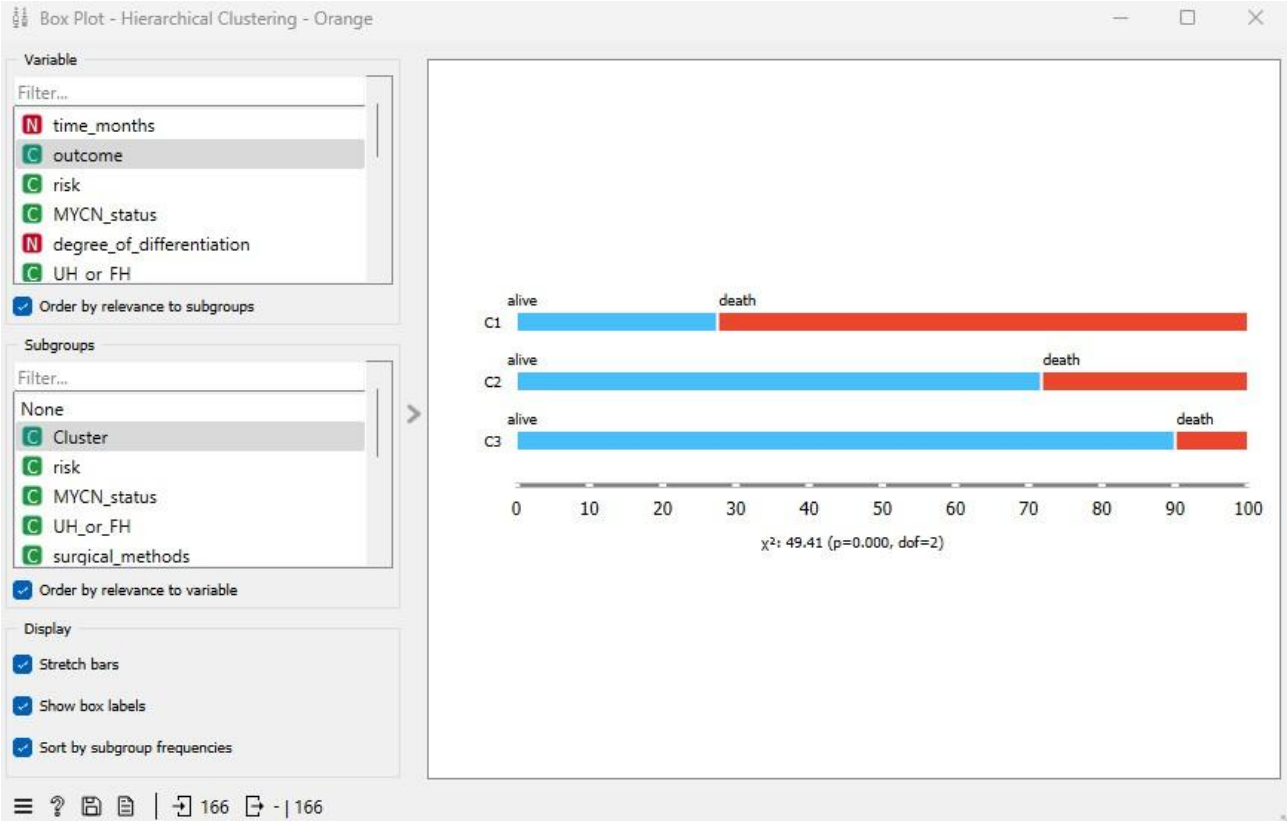
Unsupervised clustering analysis related to NEUROBLASTOMA

I standardized the variables by centering the mean at 0. This approach yielded the best **visual** division of homogeneous clusters in **Hierarchical Clustering** and resulted in the highest **Silhouette Score** for **K-Means clustering**. This effect can be attributed to the diversity of variable types: many **boolean values** combined with **continuous variables** such as **age and time in months**. With this type of standardization, the different metrics had the least possible influence on the clustering outcome.

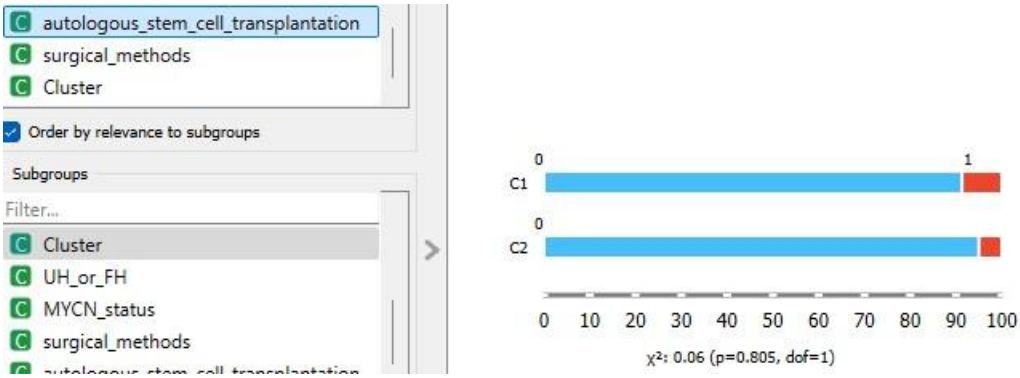


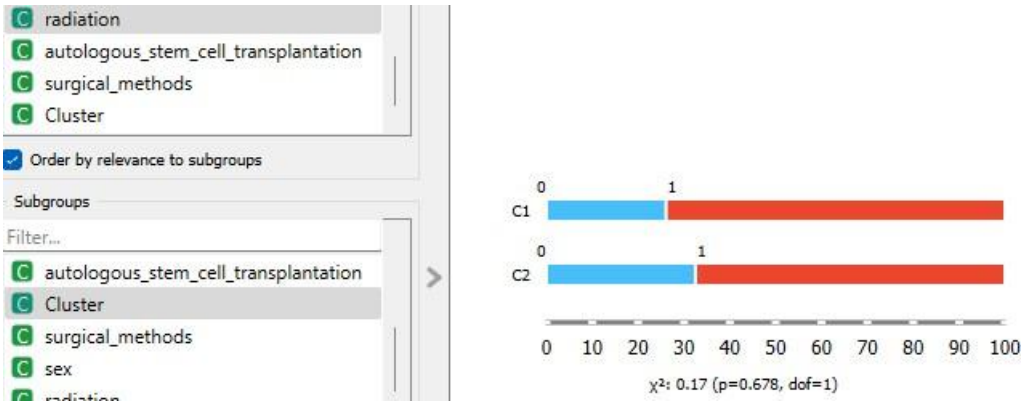
The **box plot** connected to the **hierarchical tree**, due to the significant differences in **patient outcomes** across the three identified clusters, enabled me to derive some valuable insights:

- **Cluster 1 (C1):** 80 patients, **27.5% survival rate**
- **Cluster 2 (C2):** 46 patients, **71.7% survival rate**
- **Cluster 3 (C3):** 40 patients, **90% survival rate**



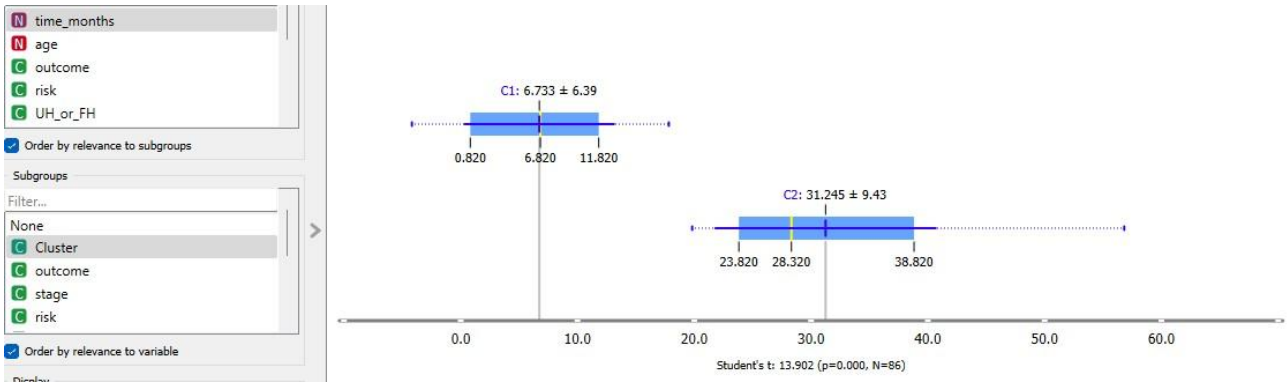
Clusters **C2 and C3** exhibit **striking similarities** regarding the **treatments administered** to patients. In both clusters, depending on the **risk level, tumor stage, and differentiation grade**, patients underwent **surgical interventions, radiotherapy, or stem cell transplantation** in almost identical proportions.



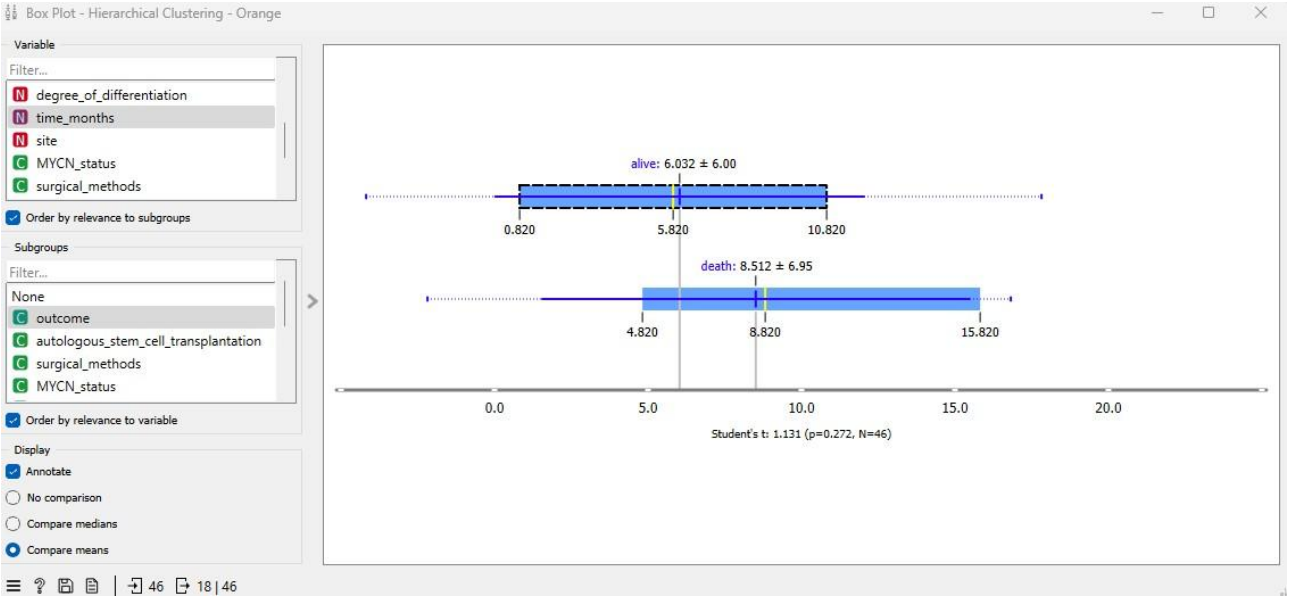


The Key Difference: Time_Months

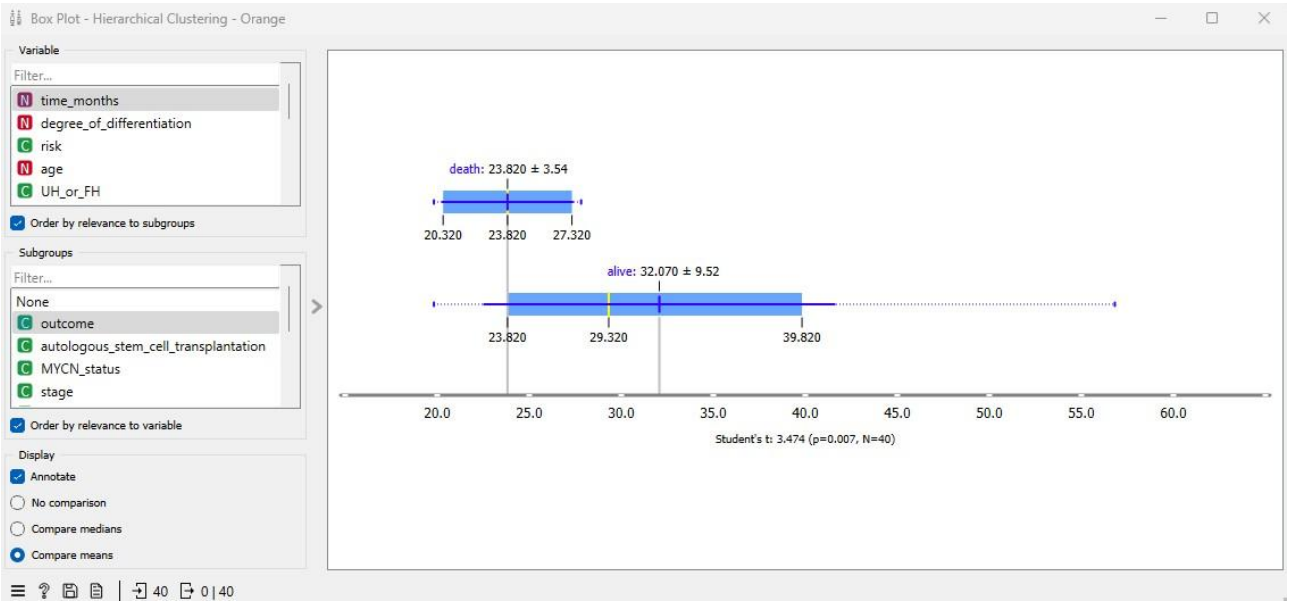
The fundamental difference between **C2 and C3** lies in the variable **"time_months."** Assuming that this variable represents the **follow-up period**, patients in **C3** were monitored for significantly longer periods than those in **C2**. The **maximum follow-up in C2 was around 18 months**, whereas in **C3, it extended up to 55 months**.



The **box plots** for **C2 and C3**, when broken down by the **"outcome"** subgroup, reveal a **considerable disparity in their medians**. This suggests that the **"time_months"** variable might directly influence the **predicted outcome**, despite similar characteristics in other clinical parameters.



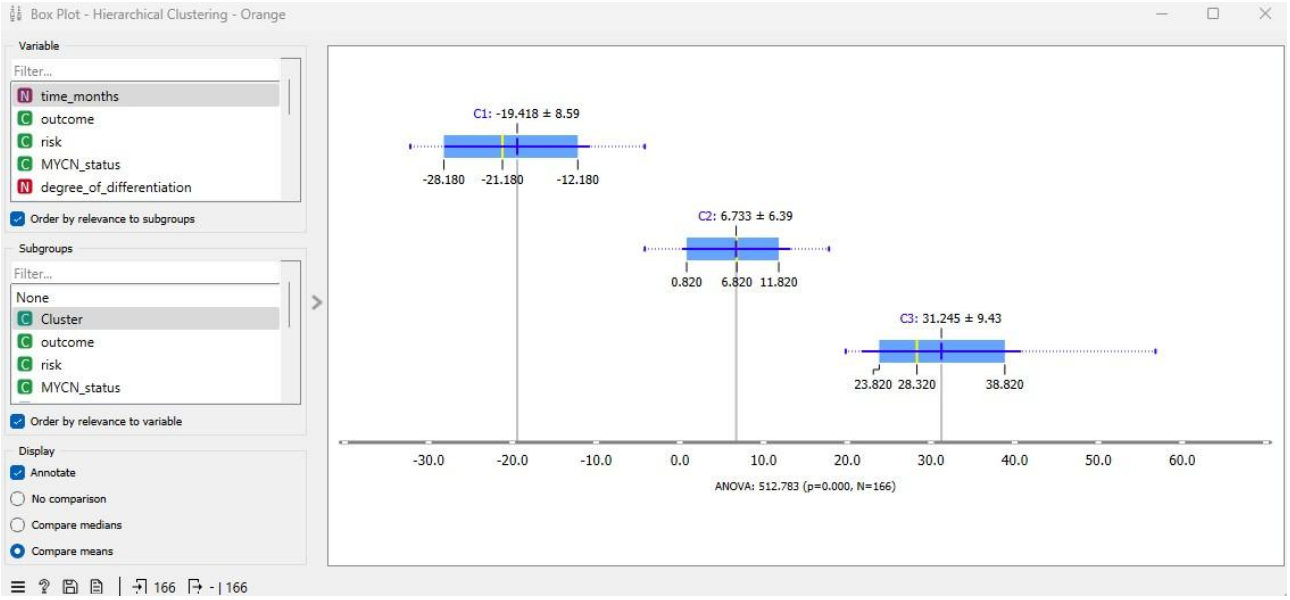
Box-plot inerente a C2

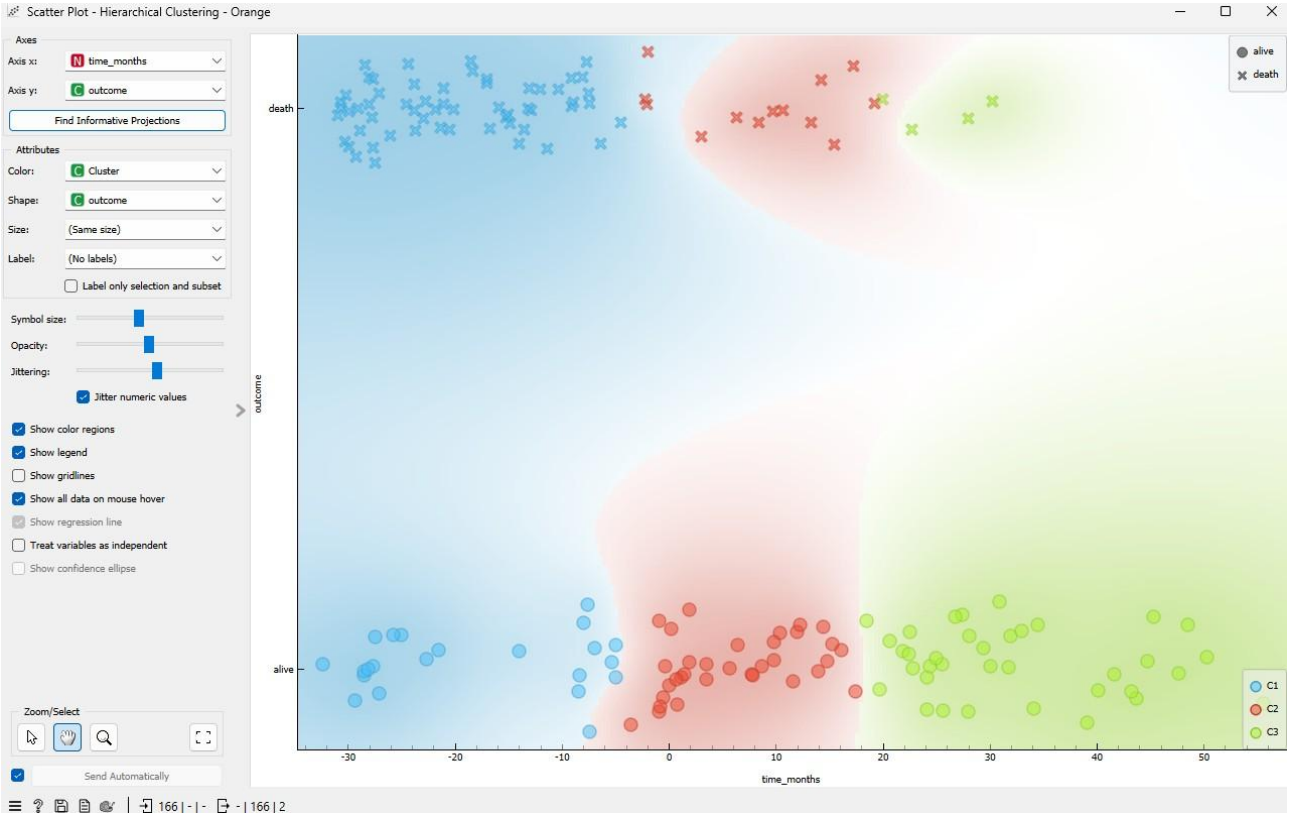


Box-plot inerente a C3

Correlation Between Outcome and Time_Months

When comparing all three clusters together and considering their respective **survival rates** (**C1: 27.5%, C2: 71.7%, C3: 90%**), the correlation between **"outcome"** and **"time_months"** becomes even more apparent.





Thus, this analysis raises the question: **Does a longer follow-up inherently contribute to better survival outcomes, or is it simply an artifact of extended monitoring?** Further statistical validation is required to determine whether "time_months" plays a causative role in survival prediction or merely reflects variations in patient observation time.